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Smolin's Natural Selection Hypothesis

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SUMMARY

Smolin has recently introduced into cosmology an intriguing natural selection hypothesis, according to which the fundamental constants evolve through mutations that take place at recollapse or black-hole singularities. He proposes that the present values of the fundamental parameters select for universes with a local maximum of black holes. The hypothesis is attractive in that it proposes an actual mechanism capable of explaining the values of the fundamental parameters. Smolin's considerations do, however, appear to contain a number of conceptual and technical flaws, which we point out in this paper.

1 INTRODUCTION

The attempt to understand why the fundamental parameters of particle physics and cosmology take on their observed values has been a long-standing, if not entirely respectable, pastime among physicists. As a general rule, such 'why' questions tend to be set aside until the easier 'how' questions are settled. In the past decade or so, physicists have apparently been more willing to address these fundamental issues. The inflationary model, which has attempted to solve the flatness and horizon problems, is one effort in this direction. The 'baby universe' scenario, which attempts to predict the value of the fundamental coupling constants through a quantum-mechanical 'sum-over-universes', approach is another. In moving from the conventional 'how' problems to the less-well charted territory of the 'why', explanations have necessarily become more speculative.

One recent attempt to provide a mechanism that accounts for the observed 'unnatural' values of the dimensionless constants has been made by Smolin in his paper 'Did the Universe Evolve?' (Smolin 1992). This unique idea represents the first attempt to incorporate the principle of natural selection into cosmology, if not into physics as a whole. It also provides a causal mechanism that is far more comprehensive than, for example, chaotic inflation in its potential ability to explain 'why things are as they are'. For this reason we feel that Smolin's proposal deserves to be taken seriously. At the same time, it appears that the mechanism, in its current stage of development, has a number of conceptual and technical flaws that throw

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some doubt on its utility. In this paper we would like to address these difficulties.

2 NATURAL SELECTION vs. THE ANTHROPIC PRINCIPLE

Smolin takes his point of departure from the famous observation that the nuclear force is roughly 10^{39} times larger than the gravitational force. According to physicists, all dimensionless numbers should be of order one, and so it becomes imperative to explain ratios such as 10^{39}. As Smolin points out, stars exist only because the gravitational force is so much smaller than the nuclear force and because the mass difference between the neutron and the proton is smaller than nuclear binding energies. In view of this, to ask why there are numbers like 10^{39} can be taken to be equivalent to the question: Why are the laws of physics and initial conditions of the universe such that the universe gives rise to stars?

Many answers have been given to this sort of question. Among physicists two points of view are currently prevalent. The first is often referred to as the Copernican principle, which states that all observed quantities in the universe are equally likely; it is then a mere accident that the fundamental parameters are such as to permit stars to exist and consequently life. The second is the anthropic principle, which states (in the broadest sense) that, were conditions any different than we observe them to be, life could not have arisen; therefore we must observe conditions to be as they are.

Smolin finds in both points of view the lack of a mechanism capable of explaining why a universe similar to our own exists. It is to remedy this lack of mechanism that Smolin has devised his natural selection hypothesis. In effect he has attempted to turn a 'why' question into a 'how' question.

Before outlining Smolin's idea, we should challenge his remarks about the anthropic principle. He writes, '...it has no explanatory power, given that we exclude explanation by final causes'. This statement is incorrect if by 'explanatory power' Smolin means what is usually meant in science, that is, 'ability to predict'. The anthropic principle comes in several non-equivalent versions, and the so-called weak anthropic principle has been used to make successful predictions. The most striking of these for some reason has failed to become widely known. It is Hoyle's 1954 realization that the last step in the Sun's triple-alpha process $^{8}Be + ^{4}He \rightarrow ^{12}C + 2\gamma$ must proceed resonantly (because ^{8}Be decays quickly) to produce sufficient carbon for life to exist (Barrow & Tipler 1986). Hoyle predicted that a ^{12}C resonance must lie at about 7.7 MeV, a prediction that was soon confirmed. He also realized that the reaction leading away from carbon, $^{12}C + ^{4}He \rightarrow ^{16}O$, could not proceed resonantly, or carbon would be depleted. Oxygen has no resonance at the fateful energy.

It is therefore not entirely correct to say that the anthropic principle is incapable of making predictions.* One might go further to argue that the utility of a scientific principle is not totally embodied in its ability to make predictions. For instance, the belief that gauge theories should exhibit certain

* See also Garrett and Coles (1992).

symmetries is not directly testable by experiment, but it has led to good theories.

Nevertheless, although the weak anthropic principle provides a 'consistency check' on any theory designed to explain the evolution of the universe (it must evolve to support life), the weak principle does not provide a unique mechanism. Smolin's proposal is an attempt to fill this gap. At the same time he avoids the Copernican extreme of assuming the structure of the universe is a mere accident.

The natural selection hypothesis consciously mimics biological evolution. Smolin assumes that a spatially closed ($k = +1$) universe comes into existence with 'natural' values of the fundamental parameters (all of order one in Planck units). Consequently, this universe lives for a Planck time, then collapses into a final singularity, which in turn produces a new universe.† During the bounce, the values of the fundamental constants are reshuffled, i.e. undergo small random changes. The daughter universe may last somewhat longer than the parent before recollapsing. The process repeats. Eventually the fundamental parameters have mutated sufficiently to produce a universe that ends in numerous final singularities or, equivalently, lasts long enough to produce numerous black holes. At this stage each singularity is assumed to give birth to a new universe, so the number of offspring now increases with each generation.

Sooner or later universes will appear whose parameters allow for the existence of stars. Because one associates stars with black holes, the universes with the most stars are expected to be those with the most black holes. If one defines $\rho(p, N)$ as the distribution of universes in the Nth generation with parameters p, then one can also define a normalized limit

$$\Delta(p) = \lim_{N\to\infty} \frac{\rho(p, N)}{\int \rho(p, N)\, dp}. \qquad (1)$$

For Smolin's hypothesis to be useful, it is necessary that $\Delta(p)$ converges in the limit $N \to \infty$. It is also necessary to assume that $\Delta(p)$ is peaked around the expected number of singularities in the universe. This is reasonable: If ρ is the number of universes in the interval dp with parameters p, then it is plausible that an increase in R, the number of black holes, will increase the number of universes with similar parameters and hence $\Delta(p)$. Finally, Smolin assumes that our universe is a typical member of the ensemble defined by $\Delta(p)$. With these assumptions, Smolin arrives at his central hypothesis:

> The parameters of particle physics are such that most changes in their values should lead to a decrease in the expected number of black holes in the universe.

As Smolin points out, to check the conjecture one does not need to know anything about $\Delta(p)$ except that it is peaked around local extrema of R, the number of black holes.

† Actually this is just a new expanding phase of the same universe. However it has become customary to refer to each such phase as a new universe. Having pointed out the error of this usage, we will from now on adopt it in order to be in line with the literature in this area.

Before turning to technical difficulties, we should point out that the sentence above, 'Because one associates stars with black holes, the universes with the most stars are expected to be those with the most black holes', is our own. Smolin does not explicitly make such a statement. Indeed, he begins by requiring stars to exist, then arrives at the hypothesis just quoted. The association of stars with black holes runs tacitly throughout this paper and appears to be a necessary part of his conjecture. Yet it is this association that will introduce numerous difficulties.

It is also appropriate at this stage to inject a note of philosophical caution. Smolin has applied the biological process of evolution through mutation and natural selection to the nonbiological world. Analogies are often useful but just as often dangerous (indeed, a theory of analogies would be useful here). The basis of medieval alchemy and astrology was the idea that the macrocosm (the universe) was governed by the same laws as the microcosm (the individual). The result of the Great Analogy, as it is sometimes called, was the identification of certain elements and parts of the body with certain planets. Today, scientists dismiss such analogies as foolish due to the confusion of two types of operating systems and the lack of obvious causal influence between them.

Although evolution is perhaps the leading paradigm in present-day biology, so far the only proposal that nonbiological systems evolve in this way is the work of Eigen and Kuppers (Kuppers 1992) applying the principle of natural selection to pre-biotic molecules arranged in hypercycles and other coevolutionary structures. Thus while it is possible the principle may enable fundamental new understanding of the foundations of physical science, it is also possible that the analogy embodied in Smolin's model is inappropriate. This is what needs to be tested.

It is then necessary, in turn, to question whether the hypothesis is testable. One of the most common arguments brought against the anthropic principle is that it is impossible actually to create another universe in which the values of the constants differ from the ones we know. Anthropic arguments therefore rely on thought experiments that assume conventional physics and mathematics to rule out other values of the fundamental parameters as incompatible with life. Smolin carries out just such thought experiments in his attempt to confirm or contradict the natural selection proposal. Although this proposal is 'theoretically' testable by mathematical and physical consistency (as is much of modern physics), it appears no more nor less amenable to experiment than the anthropic principle. This does not necessarily deny the value of the proposal, but it must be borne in mind when comparing it with competing hypotheses.

We now turn to technical criticisms.

2.1 *The $k = +1$ Assumption*

One of Smolin's assumptions is that the universes are spatially closed. In spatially closed universes, all singularities merge at the Big Crunch, so that there is only one final singularity. It might appear, then, that a $k = +1$ universe can produce only one offspring. However, black hole singularities can tunnel to other universes. All the various universes will be topologically one, but causally disconnected. Therefore, even a closed universe can produce many offspring.

Nevertheless, it appears possible to relax the $k=+1$ supposition to allow open models. In this case there would be no final universal singularities, but as long as black holes are produced, this does not seem to matter. Open universes might produce stars as they expand, leading to black hole formation and the possibility of re-expansion into a new regime.

A problem that could arise in the $k=-1$ case is that black holes might tunnel into nearby regions of the same expanding domain in the universe. Then we would be in the uncomfortable position of having the local universe domain with multiply-valued fundamental constants. However, having realised this could be a problem, it becomes clear it is equally possible in the case $k=+1$. If such tunnelling into the local neighbourhood took place, some sort of boundary would have to separate the regions with different physics, such as domain walls of the type postulated in some inflationary models and in various models with topological defects; this is awkward, but not impossible.

However, this does not seem to be a necessary consequence of the supposition $k=-1$. One should remember here that the expanding regime would not be an entire $k=-1$ universe, but rather part of one (a full universe would have infinite spatial sections and so involve an infinite amount of matter; a finite amount of matter emerging into a new expansion regime after passing through a black hole stage could only create a finite portion of such a universe). Indeed it is not clear why the value of k should be preserved during the re-expansion process envisaged; perhaps both signs would occur at random. (After all, the sign of k affects the cosmological curvature but has very little effect indeed on local gravitational collapse, and consequently on the production of black holes.)

2.2 *The Neutron-Proton Mass Difference*

In attempting to show that varying the fundamental parameters leads to a decrease in the number of black holes, Smolin focuses on the neutron-proton mass difference and asks what would happen were this changed. He first argues that were $\Delta m \equiv m_m - m_p$ negative the galaxy would consist mostly of a gas of neutrons instead of hydrogen. Neutrons, because they couple to the electromagnetic field weakly, would lose energy slowly via radiative transport and collapse into stars more slowly than hydrogen clouds. Consequently the black-hole production rate in such a universe would be very low.

We believe the assumptions of this argument are incorrect, and very likely so are the conclusions. The ratio of neutrons to protons in the early universe is given by the Boltzmann distribution:

$$\frac{n}{p} = e^{-\Delta m/T}, \tag{2}$$

with $c=k=1$. In the real universe $\Delta m = 1.29$ Mev, leading to fewer n than p. As is well-known, at about 1 second after the Big Bang, the expansion rate of the universe overtakes the weak reaction rates λ_w holding n and p in equilibrium via $n \rightleftharpoons p$. (All these rates are independent of Δm.) The crossing of timescales takes place at a 'freeze-out' temperature $T = T_f$. As a result, for $T < T_f$ the n/p ratio is frozen out of equilibrium at a value $n/p \sim 0{\cdot}19$ for Δm_{real}. Neutron decay then brings this ratio down to about 0·15, with the neutron mass fraction $X_n = n/(n+p) \sim 0{\cdot}12$ at the onset of primordial nucleosynthesis, which then processes all the available neutrons into helium. Since ^{4}He contains $2n$ and $2p$, the final helium mass fraction is $X_4 \simeq 2X_n \sim 0{\cdot}25$.

2.2.1 *$\Delta m = 0$.* If in equation (2) $\Delta m = 0$, then clearly $n/p = 1$ and the universe would consist of 100 per cent helium. Helium clouds could collapse into stars, but helium stars have short main-sequence lifetimes. Thus in a $\Delta m = 0$ universe, there would probably be no long-lived stars and no life, but more black holes than in our own universe.

This clarifies the problem alluded to earlier: Smolin really needs stars, not black holes, to produce a universe like we see around us. What's more, he needs stable, long-lived stars to produce life. In the above example, one gets many black holes but few long-lived stars, in contradiction to his hypothesis.

2.2.2 *$\Delta m < 0$.* Next consider a universe with $\Delta m < 0$. In this case equation (2) still holds but neutrons outnumber protons and nucleosynthesis is proton limited. Nevertheless, if $|\Delta m| \sim 0$, the result will once again be a universe with $X_4 \sim 1$. Only when $\Delta m < 0$ and $|\Delta m| > \Delta m_{real}$ will the universe contain $X_4 \leqslant 0{\cdot}025$ with an abundance of neutrons. However, let us consider such a model.

The first point to make is that neutrons would cool in such a universe simply because they participate in the general adiabatic expansion and so their thermal velocities would be damped out. (As an empirical piece of evidence, we note that in the real universe, neutrons froze out of thermal equilibrium at about 10^{10}°. If they did not cool, it would be impossible for deuterium to bind at 10^9° and there would be no helium.) Let us then assume that a neutron-universe would cool to some effective temperature of order 10°, at which point star formation would normally occur. We confine ourselves to the isothermal situation.

In the zero-angular momentum case, a neutron cloud collapses on a timescale determined by the competition between the cloud's gravitational energy $E_g \sim GM^2/R$ and its thermal energy $E_T \sim NkT$, where N is the number of nucleons. The gravitational force per cubic centimetre (for constant mass) goes as $1/R^5$ and the isothermal pressure gradient goes only as $1/R^4$. Thus, if a cloud is massive enough to collapse, once collapse starts, the pressure becomes negligible and collapse continues. The collapse timescale can be easily estimated by dimensional arguments to be

$$t \sim \left(\frac{m_N}{kT}\right)^{3/2} GM. \tag{3}$$

This is obviously independent of charge and so the collapse timescale is the same as that for a hydrogen cloud.

However, apart from the adiabatic cooling of neutrons, there are other considerations. We list some of these now, although it will become clear that star formation in an all-neutron universe is not a trivial problem and our conclusions are tentative.

Neutrons also cool through the coupling of their magnetic dipole moment to the electromagnetic field. This coupling is sizable, and used for measurements in laboratory experiments, but we have not attempted any calculation of cooling rate in the early universe. In addition, a hydrogen cloud, unlike a neutron cloud, contains as many electrons as protons (we assume charge conservation). When a hydrogen cloud collapses to $R \sim R_{\odot}$, depending on the temperature, electron opacity due to Thomson scattering will begin to impede radiative transport, thus greatly increasing the collapse timescale of the hydrogen cloud with respect to a neutron cloud. Finally, if we include the effect of magnetic fields, in a medium where magnetic fields are important, the sign of the magnetic field pressure is the same as the thermal pressure; magnetic fields actually inhibit collapse (see equation (6–22) in Spitzer 1968). However, this effect would be operative only in a conducting medium, e.g. a hydrogen plasma, and not for a neutron cloud.

The above considerations all seem to indicate that, for the zero-angular momentum case, at least, Smolin's conclusion is probably reversed: a neutron-dominated universe should have at least as many black holes as a proton-dominated universe.

The case of non-zero angular momentum is yet more difficult and our conclusions are even more tentative. For our purposes, the important question is how the timescales for angular-momentum loss compare between neutron clouds and hydrogen clouds.

Although magnetic fields would probably cause angular momentum loss in a hydrogen cloud more quickly than they could in a neutron cloud (again, the neutrons do couple through their magnetic dipole moment), other considerations indicate that star-formation timescales are not highly charge-dependent: molecular clouds where star formation takes place are neutral or not highly ionized. Thus, the scenarios for a neutron-cloud collapse and a hydrogen cloud collapse should be roughly similar. Furthermore, an important angular momentum loss mechanism appears to be shedding; most stars form binary or multiple systems to change rotational angular momentum into orbital angular momentum, or else those parts of the initial cloud with high angular momentum do not collapse into stars. (Recall that Jupiter contains virtually all the angular momentum of the solar system.) Hence, the parts of the cloud that collapse are low angular momentum. Indeed one of the few points of agreement among researchers seems to be that the initial stage of collapse into cores takes place on a hydrodynamic timescale; the angular momentum is not the dominant factor (Strom 1976; Cameron 1988; Longair 1989).

Taking all these factors into consideration, there seems to be little reason at this stage of investigation to think that a $\Delta m < 0$ universe would contain fewer black holes than our own, and several plausible reasons to believe it would contain more.

2.2.3 *$\Delta m \gg 0$.* Turning to the opposite extreme, Δm positive and greater than the binding energy of helium, Smolin notes that no stable elements heavier than hydrogen would exist. In such a universe there would be no stellar evolution and no supernovae, again prohibiting the development of life.

In this hypothetical universe, Smolin assumes that any cloud of hydrogen gas of mass larger than the Landau-Chandrasekhar limit collapses into a black hole. (Note the apparent contradiction with the assumption that neutron clouds do not collapse into black holes.) Such a universe would clearly have many black holes. In our own universe, however, massive stars do undergo mass loss through a variety of mechanisms (in particular supernovae) and the ejected mass can be reprocessed into stars and conceivably black holes. With a simple recycling model Smolin finds, nevertheless, that a pure hydrogen universe contains more black holes than the present one. This is not terribly surprising; to change this conclusion practically all mass that is lost by stars would have to be processed into black holes.

Smolin's recycling model, however, depends on the assumption that the stellar birth-rate function is independent of Δm. He goes on to argue that the presence of complex chemicals and dust grains plays a major role in star formation, and so the assumption is not correct. Smolin does not provide a new estimate for the stellar birth-rate function or a new estimate for the number of black holes. Although in principle his point is correct, we repeat that simulations of star formation appear to indicate that in the initial stage, at least, gravity dominates, regardless of other assumptions. Furthermore, it would appear that the addition of complex molecules and dust could only increase the opacity, in which case collapse would proceed more slowly, decreasing the black-hole birth rate.

Again it appears that Smolin's conclusion is probably reversed.

3 INFLATION AND THE QUESTION OF PRIMORDIAL BLACK HOLES

The difficulty that surfaced in the previous section and which will become more apparent is that there are more ways to produce black holes than long-lived stars, and so a large population of black holes is not equivalent to a large population of stable stars. This problem intrudes into the attempt to discover whether natural selection leads to universes that undergo inflation.

Smolin focuses his attention on λ, the self-coupling of the scalar field (governing the size of the inflationary potential; Mazenko, Unruh & Wald 1985). In new-inflationary models λ must be fine-tuned: if λ is too large, inflation does not occur due to the objections of Mazenko *et al.* (1985), or it produces fluctuations $\delta\rho/\rho$ much larger than those observed in the microwave background (leading to perfectly possible universes, but not the one we live in). On the other hand, for a universe to produce black holes, $\delta\rho/\rho$ cannot be too small; and since $\delta\rho/\rho$ grows with λ, λ cannot be too small either.

For natural selection to work, therefore, in a universe with inflation, one needs to optimize λ such that it balances these competing tendencies. Smolin produces a simple model in which for $\lambda > \lambda_{crit}$ no inflation takes place and the number of new universes is $R = 1$. For $\lambda \leqslant \lambda_{crit}$ inflation takes place and produces a large number of black holes $R = K(Vol)G(\delta\rho/\rho)$, where K is a monotonically increasing function of the universe's volume and G is a monotonically increasing function of $\delta\rho/\rho$. Since $\delta\rho/\rho$ is a monotonically increasing function of λ, G is as well.

For his toy model, Smolin takes K to be a step function. That is, for $\lambda > \lambda_{crit}$, inflation does not take place and $K = 0$. In this case, he sets $R = 1$.

But for $\lambda \leqslant \lambda_{crit}$, $K =$ constant $\gg 1$. For this simple model, R is clearly peaked at $\lambda = \lambda_{crit}$, but for more general forms of K and G, R could be peaked anywhere, conceivably in a region that does not allow inflation. Now there is no real reason to reject non-inflationary models, from the viewpoint of the Smolin programme, unless one supposes that inflationary universe models do produce many more black holes than non-inflationary ones. There is, however, a good reason to believe the opposite: that reason is primordial black holes.

From the point of view of tunnelling to new universes, primordial black holes would serve just as well as stellar remnants. Thus, natural selection should produce universes with large numbers of primordial black holes. Yet, the 'generic' inflationary model is exactly the one that suppresses primordial black hole formation by damping out inhomogeneities. At least any inflationary model that produces fluctuations compatible with the COBE limits will not produce primordial black holes. (One could presumably fine-tune λ to produce an inflationary universe with many black holes but this would not resemble our own in terms of the microwave background.) If, therefore, one wants to select for 'generic' inflation (which one may or may not wish to do), one apparently needs to exclude primordial black holes from the conjecture. We return to this point shortly.

4 NATURAL SELECTION AND PRIMORDIAL NUCLEOSYNTHESIS

Smolin also asks whether natural selection could have operated through processes during the epoch of primordial nucleosynthesis. In his scenario, the energy released by strong nuclear reactions increases the kinetic energy of the universe and hence the expansion rate. The lifetime of the universe t_u before final collapse is also increased in this way. Smolin requires that t_u be large enough to allow the formation of black holes. Assuming that the entire kinetic energy of the universe is due to nuclear reactions, he finds that $t_u \sim 100 t_{Hubble}$ in order to bring the energy release per baryon down to ~ 1 MeV, a value low enough to be compatible with the known spectrum of particle masses. In other words, he is again selecting for Δm. Smolin then argues that this scenario is within the realm of possibility since current thinking allows $t_u \gg t_{Hubble}$.

This model has several fatal flaws. First, in the conventional Big Bang picture, the universe is radiation dominated during nucleosynthesis, so the baryons can have no dynamical effect on its evolution. Therefore Smolin's scenario takes place in a cold or tepid Big Bang. Such models have not been taken seriously for a long time and, in the wake of the COBE observations, they can almost certainly be pronounced dead.

Moreover, it is difficult to imagine a cold Big Bang that also selects for inflation, since GUTs are only operative at high temperatures. One would require the universe to begin hot, then cool off before nucleosynthesis. Finally, as discussed by Carr (1985), cold and tepid Big Bangs are prolific generators of primordial black holes. Intuitively this makes sense because a cold Big Bang's soft equation of state $p \sim 0$ makes it easier for matter to collapse under gravity than in a radiation-dominated phase with a hard equation of state.

If Smolin wishes to argue that the cold Big Bang is the real universe, then indeed it perhaps contains a measurable fraction of its mass in primordial black holes, but these are not stars. Once again it seems more sensible to modify the natural selection hypothesis to exclude primordial black holes. Then one selects against cold Big Bang models and, in selecting for inflation, one automatically mitigates against primordial black holes. However, this raises the problem mentioned earlier: primordial black holes are presumably just as capable as any others of tunnelling into new universes. Indeed, the more universes created, the more are likely to have similar parameters. It is therefore difficult to see why $\Delta(p)$ should be peaked around maxima of R(stellar remnant holes) and not around maxima of R(all holes).

Many of the same remarks apply to very massive objects and supermassive objects, which could collapse into black holes directly without making stars. Such objects are also expected to be more common in cold or tepid Big Bangs. In all respects, in view of current understanding, it is certainly better to choose a theory that prefers a hot big bang to a cold or tepid one.

5 OTHER CONSTRAINTS

A more general difficulty with Smolin's paper is that he considers only a few parameters. There is no obvious reason why evolution should not select for all the significant constants, including, for example, the fine-structure constant α; indeed if we take the theory seriously, this is what we would expect.

Were α lowered slightly from its present value, coulomb repulsion between two protons would be reduced so that ^{2}He, the diproton, would be bound. In the early universe protons would form diprotons on a strong-interaction timescale. All hydrogen would therefore be processed into ^{2}He, resulting in a universe without water, without long-lived stars and consequently with a greater black-hole production rate than our own.

Once again we see that a universe with large numbers of black holes is not equivalent to a universe with a large stellar population. One can investigate this problem further by considering what would seem to be the most obvious way to increase the number of black holes: lower the Landau–Chandrasekhar mass

$$M_{LC} \sim \alpha_g^{-3/2} m_N, \tag{4}$$

where the gravitational fine structure constant $\alpha_g \equiv G m_N^2/\hbar c$ and m_N is the nucleon mass.

If α_g is increased, M_{LC} will go down. An increase in the gravitational force will result in an increase in the rate of gas cloud contraction. To reach stellar equilibrium will require higher temperatures and pressures, resulting in faster nuclear burning and fewer long-lived stars, but more black holes.

However, the matter may not be so simple. If one accepts the arguments of Rees (1976), then star formation depends on the fragmentation of the initial gas clouds. The fragment mass has exactly the same dependence on α_g as does M_{LC}. In this case, one would lower M_{LC}, but the fragment mass would go down by the same amount. If the mass spectrum of clouds remained unchanged, then presumably the same proportion would form black holes as before. Consequently, if this argument is correct, changing α_g would have little or no effect on the number of black holes.

The above examples show in one case that it is very easy to change parameters in a direction that make fewer stars but more black holes, and in another case, that the number of black holes may be insensitive to a parameter change. Smolin's scenario, however, requires that changing parameters in either direction decreases the number of black holes. But, clearly, raising α or M_{LC} will work in the opposite direction than just described (or in the latter case perhaps do nothing). In general, it is difficult to think of any parameter change that works in only one direction. This basic flaw in Smolin's scenario is actually quite similar to that in the famous Collins and Hawking (1973) anthropic argument for the isotropy of the universe. They claimed that the universe must be flat and isotropic; otherwise perturbations would not grow fast enough to produce galaxies, or the universe would recollapse. However, they chose their initial amplitudes to give galaxies by the present time in a flat universe. One could alter these initial amplitudes to produce no galaxies, or entirely black holes.

6 CONCLUSIONS

The Smolin argument is a very interesting one, and opens up new vistas for explanation in cosmology. As it stands it has several problems and obscurities, some of which we have pointed out in this paper. Apart from the technical problems of making parameter changes operate unidirectionally – and in the correct direction – the primary requirement at this stage is a mechanism to exclude primordial black holes from the proposal.

One possibility is a preferential tunnelling rate into a new expansion phase that depends on the mass of the black hole that has collapsed. Whether or not such a dependency is possible depends on the physics of the tunnelling process, which is at present highly speculative. This proposal has the obvious flaw that primordial black holes can also be very massive, but it would at least exclude microscopic black holes.

Despite such problems, in view of the power of the process of natural selection as a mechanism for creating apparent design (see e.g. Dawkins 1988), the programme is certainly worth pursuing in the broad context outlined by Smolin.

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